THE DIMER ENTROPY OF THE KING'S GRAPH

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Abstract. We compute the entropy per site of the perfect matchings (dimer coverings) of the king's graph, the square lattice with both diagonals; equivalently, of the domicule tilings of rectangular boards counted in OEIS A239264 and A239273. Row transfer matrices on cylinders of circumference up to 25, split by a conserved parity, give

$$s = 0.5655119110413253 \pm 6 \cdot 10^{-16},$$

with an error bar supported by several extrapolation models, backtests and an extended-precision check of the rounding. We prove $0.545343749727 \le s \le 0.586216605404$ with computer-assisted Collatz–Wielandt certificates; the upper bound uses a Fibonacci bound for the matchings across a cut. We also give the surface free energy of a free edge, $\sigma = 0.242017935766251 \pm 9 \cdot 10^{-15}$, and the exponential decay of the splitting between the two parity sectors, with decay length $\xi \approx 1.23$. Finally we state the conjecture that $q \mapsto Z(2q, m)$ is log-convex for every strip width m. It would raise the rigorous lower bound to $s \ge 0.565511911035$; it holds in all 1451 cases we checked, but it fails on the triangular lattice and when the two diagonals carry clearly different weights.

1. Introduction

A *domicule* is a pair of cells of the square grid that share a side or a corner. A domicule tiling of a $K \times L$ board is a partition of its cells into domicules; two diagonal domicules may cross inside a unit square, so the 2×2 board has 3 tilings. Equivalently, it is a perfect matching of the *king's graph* on the board, the graph whose edges join cells at chessboard king distance 1. The numbers $Z(K, L)$ of such tilings form the array A239264 of the OEIS [13], and $Z(2n, 2n)$ is A239273 [12]; related entries such as A239265, A239268 and A243510 [9, 10, 11] record further exact counts. None of these entries gives the growth constant, i.e. the entropy per site

$$s = \lim_{K,L\to\infty} \frac{\log Z(K, L)}{KL}.$$

For dominoes (the square lattice) s is known exactly, G/π with Catalan's constant G, by the Pfaffian method of Kasteleyn and of Temperley and Fisher [7, 16]. Kasteleyn's method needs a Pfaffian orientation, which exists for planar graphs, and the triangular lattice (one diagonal per square) is planar and was solved in the same way [4]. The king's graph is not planar: the 5×5 board already has $72 > 3 \cdot 25 - 6$ edges. No Pfaffian or determinantal solution of the king's dimer model is known. Sandvik and Moessner [15], and before them Sandvik [14], studied exactly this model (dimers on the nearest- and next-nearest-neighbour bonds of the square lattice) by Monte Carlo simulation; they note that it is not solvable by Kasteleyn's method and do not give its entropy. Wildeboer et al. [17] solve a different, constrained dimer model on lattices with crossed plaquettes. Finch's errata [5, §5.12] list a constant $1.34254\ldots1.34265$ for the square lattice with "crossed diagonal bonds"; it belongs to hard squares (independent sets of the king's graph), not to dimers. Series expansions for dimer and monomer–dimer problems go back to Nagle [8] and Gaunt [6]; we have only seen their abstracts, which do not mention the king lattice. To our knowledge no numerical value of s has been published.

Date: Draft, October 2026.

Results.

(1) $s = 0.5655119110413253 \pm 6 \cdot 10^{-16}$ (§3), i.e. $2s = 1.131023822082651$ per dimer and $e^s = 1.760348694041978$. The error bar is not rigorous.
(2) Rigorously, $0.545343749727 \le s \le 0.586216605404$ (Theorem 4.1).
(3) The surface free energy of a free edge is $\sigma = 0.242017935766251 \pm 9 \cdot 10^{-15}$ (§5).
(4) On even cylinders the two parity sectors split by $\Delta_w \approx e^{-w/\xi}$, $\xi \approx 1.23$; all finite-size corrections are exponentially small (§6), in contrast with the critical square lattice.
(5) The conjecture (L) that $q \mapsto Z(2q, m)$ is log-convex would give the rigorous bound $s \ge 0.565511911035$ (eleven correct digits). We give the evidence, the obstructions to the standard proofs and two counterexamples in neighbouring models (§8).

The same code reproduces the exactly known entropies of the square and triangular lattices (§7). All scripts, outputs and the map from every number in this paper to the file it comes from are in the repository (§9).

2. Model, transfer matrix and parity sectors

2.1. **Definitions.** The king's graph has vertex set $\mathbb{Z}^2$, and $(r, c) \sim (r', c')$ if $\max(|r-r'|, |c-c'|) = 1$. The board $B(K, L) = \{0, \dots, K-1\} \times \{0, \dots, L-1\}$ has K rows and L columns; $Z(K, L)$ is the number of perfect matchings of the induced subgraph ($Z = 0$ if KL is odd), and $Z(K, L) = Z(L, K)$. The cylinder of circumference $w \ge 3$ identifies column c with $c + w$; its count is $Z_{\mathrm{cyl}}(K, w)$. Let

$$\bar{s} = \limsup \frac{\log Z(K, L)}{KL}, \qquad \underline{s} = \liminf \frac{\log Z(K, L)}{KL},$$

over $K, L \to \infty$ with KL even. That $\underline{s} = \bar{s}$ is standard for dimer models on $\mathbb{Z}^2$-periodic graphs (a sub/superadditivity argument); Theorem 4.1 bounds $\underline{s}$ and $\bar{s}$ separately and does not use it.

For a strip of width k, stacking boards gives $Z(2i + 2j, k) \ge Z(2i, k) Z(2j, k)$, so by Fekete's lemma [3] the limit

$$\lambda_k^2 = \lim_{j \to \infty} Z(2j, k)^{1/j} = \sup_j Z(2j, k)^{1/j} \tag{1}$$

exists. We call λ_k the growth rate of the free strip and write $a_k = \log \lambda_k$. Placing strips side by side gives $\lambda_{k_1 + k_2} \ge \lambda_{k_1} \lambda_{k_2}$, so $a_k / k \to s^* = \sup_k a_k / k$.

2.2. **Transfer matrix.** We scan a strip of width w cell by cell in row-major order. A cell is either already covered, or it is matched to one of its later neighbours: right $(r, c+1)$, down-left $(r+1, c-1)$, down $(r+1, c)$ or down-right $(r+1, c+1)$, if that neighbour exists and is free. The state records the occupancy of the next $w + 2$ cells; on the cylinder the neighbours are taken modulo w and one extra bit records the cell $(r+1, w-1)$ claimed by $(r, 0)$ through the wrap-around down-left edge. A row is w cell steps, $T = T_{w-1} \cdots T_0$, and

$$Z(K, w) = e_0^\top T^K e_0,$$

with e_0 the empty state. Between rows only the w bits of the next row can be set, so T acts on 2^w states: the set $A \subseteq \{0, \dots, w-1\}$ of cells of the next row already covered from the current one. The implementation uses reshaped array views instead of index arrays and is bit-identical to a direct one.

2.3. **Parity sectors.** Let A_r be the set of cells of row r covered from row $r-1$ and h_r the number of dimers inside row r. Counting the cells of row r gives $w = |A_r| + 2h_r + |A_{r+1}|$, so

$$|A_{r+1}| \equiv w - |A_r| \pmod 2.$$

Hence T^2 preserves the parity p of $|A|$; for even w so does T, for odd w the map T exchanges the two sectors and T^2 has the same nonzero spectrum on both. We write $\lambda_w^{(p)}$ for the square root of the leading eigenvalue of T^2 on sector p of the cylinder, and $f_w^{(p)} = \log \lambda_w^{(p)}/w$. On the cylinder $|A|$ is the number of dimers crossing a closed loop around the axis, and for even w its parity is conserved: this is the topological $\mathbb{Z}_2$ sector familiar from dimer models on non-bipartite lattices. For even w the two sector values differ, and they bracket s in all our data: $f_w^{(0)} > s > f_w^{(1)}$.

For free strips only the sector of e_0 enters Z, and the spectral radius of T is not the growth rate λ_w in general: for $w = 6, 10, 14, 18$ the other sector has the larger eigenvalue (e.g. $w = 6$: $\log\lambda = 2.909568954808860$ there, against $a_6 = 2.909403518708946$).

2.4. **Computation.** We use float64 power iteration on T^2 restricted to one sector, from a random positive start vector (fixed seed), normalised every step, stopped when the estimate of $\log\lambda$ changes by less than $4 \cdot 10^{-15}$ for three consecutive iterations (19 iterations for $w \geq 20$). Afterwards the Collatz–Wielandt bracket $\min_i (T^2 v)_i/v_i \leq \lambda^2 \leq \max_i (T^2 v)_i/v_i$ over the support of v is evaluated; for $w = 20, \ldots, 25$ it closes to the last digit of $\log\lambda$. Widths $w \leq 19$ take minutes; $w = 25$ (one sector suffices for odd w) works on vectors of 2^{28} doubles and needs 8.4 GiB and 22.5 minutes on four cores.

3. The value of s

3.1. **Data.** Table 1 lists the odd cylinders. The values f_w increase to s; the differences $d_w = f_w - f_{w-2}$ decay by a ratio $r_w = d_{w-2}/d_w$ that drifts from 13.4 to 9.99. The geometric (Aitken) estimate $s_w = f_w + d_w/(r_w - 1)$ settles at 0.5655119110413253. For even w the sector values bracket s:

$w = 18$: [0.5655119060934878, 0.5655119160322841],
$w = 20$: [0.5655119102001319, 0.5655119118867463],
$w = 22$: [0.5655119108963251, 0.5655119111867462],
$w = 24$: [0.5655119110160359, 0.5655119110666571], midpoint 0.5655119110413465,

and the midpoints converge to s at the same rate as the odd widths, from independent data.

Table 1. Odd cylinders: $f_w = \log\lambda_w/w$, $d_w = f_w - f_{w-2}$, $r_w = d_{w-2}/d_w$ and the geometric extrapolation s_w (`out_estimate.txt`).

w	f_w	d_w	r_w	s_w
13	0.5655119032044848	$7.943 \cdot 10^{-8}$	11.68	0.5655119106414456
15	0.5655119103137916	$7.109 \cdot 10^{-9}$	11.172	0.5655119110126792
17	0.5655119109720268	$6.582 \cdot 10^{-10}$	10.801	0.5655119110391899
19	0.5655119110345892	$6.256 \cdot 10^{-11}$	10.521	0.56551191104116
21	0.5655119110406597	$6.071 \cdot 10^{-12}$	10.306	0.5655119110413121
23	0.5655119110412587	$5.989 \cdot 10^{-13}$	10.136	0.5655119110413242
25	0.5655119110413186	$5.994 \cdot 10^{-14}$	9.992	0.5655119110413253

3.2. **Extrapolation models and error bar.** The tail $s - f_W = \sum_{j\ge1} d_{W+2j}$ was extrapolated with seven models, each fitted on the last few widths: geometric (G, constant ratio r_W); $d_w = Ae^{-w/\xi}w^b$ through three points (PE) or by least squares on six (PE6); $r_w = r_\infty + c/w$ (R1) and $r_w = r_\infty + c/w + e/w^2$ (R2); two exponentials by Prony's method (2E); and two variants of Wynn's ε algorithm (WY). Each model was rerun with the data cut at $W = 19, 21, 23$ (backtest), and the same models were applied to the even-width midpoints. Table 2 shows the result.

TABLE 2. Estimates of s by model: from the odd widths up to $W = 25$, from the even midpoints up to $W = 24$, and the backtest change $|s(23) - s(25)|$ on the odd widths (`out_errbar.txt`).

model	odd, $W = 25$	midpoints, $W = 24$	backtest
G	0.5655119110413253	0.5655119110413256	$1.05 \cdot 10^{-15}$
PE	0.5655119110413254	0.5655119110413253	$3.13 \cdot 10^{-17}$
PE6	0.5655119110413254	0.5655119110413253	$2.67 \cdot 10^{-17}$
R1	0.5655119110413254	0.5655119110413253	$4.2 \cdot 10^{-18}$
R2	0.5655119110413254	0.5655119110413253	$5.14 \cdot 10^{-17}$
2E	0.5655119110413254	0.5655119110413253	$1.12 \cdot 10^{-16}$
WY	0.5655119110413254	0.5655119110413253	$8.8 \cdot 10^{-17}$

The geometric model is the only one that fails its backtest (it ignores the drift of r_w, which the PE fit attributes to a power prefactor: $\xi_{\text{odd}} = 0.9324$, $b = -1.880$). The models that pass their backtest (change $< 5 \cdot 10^{-16}$) on either data set give

$$s \in [0.5655119110413252,\ 0.56551191104132547],$$

with centre 0.56551191104132534 and half-width $1.3 \cdot 10^{-16}$.

Rounding. We recomputed the odd cylinders $w = 15, 17, \dots, 25$ in x87 extended precision (64-bit mantissa). The float64 values differ by at most $4.1 \cdot 10^{-17}$ per site, and the extrapolation has sensitivity $\sum_w |\partial s_{\text{G}}/\partial f_w| = 1.49$. On the extended-precision values alone the models give 0.56551191104132531–2.

Cross-checks of the float64 eigenvalues. Exact integer ratios $Z_{\text{cyl}}(K+2, w)/Z_{\text{cyl}}(K, w)$ for $w = 9, 11, 12, 13$ agree with the power iteration to $4.1 \cdot 10^{-16}$ in $\log\lambda$, and ARPACK on the full $w = 20$ transfer reproduces both sector values to $1.8 \cdot 10^{-15}$.

We take three times the sum of the model half-width and the propagated rounding:

$$\boxed{s = 0.5655119110413253 \pm 6 \cdot 10^{-16}}$$

This is an estimate, not a bound. For comparison, the exact counts of the square boards up to 20×20 alone give about six digits (a fit $\log Z(L, L) = sL^2 + aL + b$ on $L = 16, 18, 20$ gives 0.565512081672).

4. RIGOROUS BOUNDS

Theorem 4.1 (computer-assisted)**.** $0.545343749727 \le \underline{s} \le \bar{s} \le 0.586216605404$.

For comparison, the Kahn–Lovász bound (the number of perfect matchings of a graph is at most $\prod_v (d_v!)^{1/(2d_v)}$; unpublished by its authors, proved in [1]), with all degrees $d_v \le 8$, gives $\bar{s} \le \log(8!)/16 = 0.662787681422$.

Both bounds of Theorem 4.1 come from strips of width $w = 24$ and a numerical certificate whose validity does not depend on how it was found. Table 3 shows how they improve with

w. Since $\log \lambda_w/w = s + f(w)/w$ with $f(w) \to -2\sigma$ (§5), the lower bound approaches s like $2\sigma/w = 0.484/w$; the upper bound does like $0.497/w$.

TABLE 3. Certified bounds from strips of width w (`out_bounds.txt`).

w	lower bound on $\underline{s}$	upper bound on $\bar{s}$
4	0.445893077494	0.689765394232
8	0.505014525148	0.627626147026
12	0.525175645897	0.606921301094
16	0.53525966962	0.596568952598
20	0.54131011747	0.590357544276
24	0.545343749727	0.586216605404

4.1. **Lower bound.**

Lemma 4.2 (superadditivity)**.** *For even K and $L = kw+r$ with $0 \le r < w$: $Z(K, L) \ge Z(K, w)^k$.*

Proof. Perfect matchings of k strips $B(K, w)$ placed side by side, completed by vertical dominoes in the last r columns (K is even), are distinct perfect matchings of $B(K, L)$. □

Lemma 4.3 (Collatz–Wielandt, lower)**.** *Let $A = T^2$ be the free-strip transfer of width w, so that $Z(2j, w) = (A^j e_0)_0$. Let $v \ge 0$ with $v_0 > 0$, $\operatorname{supp} v \subseteq \operatorname{supp} A^n e_0$ for some n, and $Av \ge c\,v$ componentwise. Then $Z(2(n+j), w) \ge c^j v_0/C'$ for all j, where $C' = \max_{i \in \operatorname{supp} v} v_i/(A^n e_0)_i < \infty$. In particular $\lambda_w^2 \ge c$.*

Proof. $v \le C' A^n e_0$ componentwise and $A \ge 0$, so $Z(2(n+j), w) = (A^j A^n e_0)_0 \ge (A^j v)_0/C' \ge c^j v_0/C'$. □

Proof of the lower bound. By Lemma 4.2, for even K,

$$\frac{\log Z(K, L)}{KL} \ge \frac{\lfloor L/w \rfloor}{L} \frac{\log Z(K, w)}{K} \longrightarrow \frac{\log \lambda_w}{w} \qquad (K, L \to \infty);$$

for odd K (then L is even) exchange the roles of the sides. So $\underline{s} \ge a_w/w$ for every w, and $\underline{s} \ge s^*$. The certificate of §4.3 gives $a_{24}/24 \ge \frac{1}{48} \log c_{24} \ge 0.545343749727$. □

4.2. **Upper bound.** Cut $B(K, kw)$ into k strips $S^{(1)}, \dots, S^{(k)}$ of width $w \ge 2$ (columns $jw, \dots, jw + w - 1$). King edges change the column by at most one, so every edge of a perfect matching X lies inside a strip or joins column $jw - 1$ to column jw (an edge across cut j). Let A_j be the rows of the cells of column $jw - 1$ matched across cut j (right side of $S^{(j)}$), B_{j+1} the rows of the cells of column jw matched across cut j (left side of $S^{(j+1)}$), and $B_1 = A_k = \emptyset$. Let $N(B, A)$ be the number of perfect matchings of

$$B(K, w) \setminus \big(\{(b, 0) : b \in B\} \cup \{(a, w-1) : a \in A\}\big),$$

and $M(A, B)$ the number of perfect matchings of the bipartite graph on $A \sqcup B$ with $a \sim b$ iff $|a - b| \le 1$.

Lemma 4.4 (cut identity)**.** *$Z(K, kw) = \sum \prod_{j=1}^{k} N(B_j, A_j) \prod_{j=1}^{k-1} M(A_j, B_{j+1})$, the sum over all $A_1, B_2, A_2, \dots, B_k$ with $B_1 = A_k = \emptyset$.*

Proof. Since $w \ge 2$ the sets are well defined. The restriction of X to strip j is a perfect matching of $S^{(j)} \setminus (B_j \cup A_j)$, and its restriction to cut j a perfect matching between A_j and B_{j+1}; conversely every such tuple of pieces is a perfect matching of $B(K, kw)$. □

Lemma 4.5 (crossing matchings)**.** *If $|A| = |B| = n$ then $M(A,B) \le F_{n+1} \le \varphi^n$, where $F_1 = F_2 = 1$ are the Fibonacci numbers and $\varphi = (1+\sqrt{5})/2$; $M(A,B) = 0$ if $|A| \ne |B|$.*

Proof. Let $a_1 < \cdots < a_n$, $b_1 < \cdots < b_n$ and $M(n) = \max M(A,B)$. *a_1 is matched to b_1 or b_2*: if $a_1 \sim b_j$ with $j \ge 3$, then $b_1 \le b_3 - 2 \le a_1 - 1$, while every $a \ne a_1$ has $a \ge a_1 + 1 \ge b_1 + 2$, so b_1 has no partner. *If $a_1 \sim b_2$*, then $b_1 \le b_2 - 1 \le a_1$ needs a partner $a_i \ge a_1 + 1$ with $a_i \le b_1 + 1 \le a_1 + 1$, so $a_i = a_1 + 1$, $b_1 = a_1$, and the pair $a_2 b_1$ is forced. Thus every matching contains $a_1 b_1$ (the rest is a matching of $n-1$ against $n-1$ cells) or both $a_1 b_2$ and $a_2 b_1$ (the rest is $n-2$ against $n-2$), so $M(n) \le M(n-1) + M(n-2)$ with $M(0) = M(1) = 1$, i.e. $M(n) \le F_{n+1}$. Finally $F_{n+1} \le \varphi^n$ by induction. □

The bound is attained by $A = B$ an interval. Put $z = \sqrt{\varphi}$ and

$$W(K,w) = \sum_{A,B} N(B,A)\, z^{|A|+|B|},$$

the weighted number of matchings of a strip in which a cell of the first or last column may be "covered from outside" at weight z.

Lemma 4.6 (strip inequality)**.** $Z(K, kw) \le W(K,w)^k$.

Proof. In Lemma 4.4 replace $M(A_j, B_{j+1})$ by $\varphi^n = z^{|A_j|} z^{|B_{j+1}|}$ (Lemma 4.5). The factor $z^{|A_j|}$ belongs to strip j and $z^{|B_{j+1}|}$ to strip $j+1$, so each summand is at most $\prod_j N(B_j, A_j) z^{|B_j|+|A_j|}$. Dropping the constraints $|A_j| = |B_{j+1}|$ and $B_1 = A_k = \emptyset$ adds only nonnegative terms, and the sum factorises over the strips. □

The transfer of §2 computes W when a cell in column 0 or $w-1$ may also be covered from outside at weight z: $W(K,w) = e_0^\top T_z^K e_0$, $T_z \ge 0$.

Lemma 4.7 (Collatz–Wielandt, upper)**.** *If $A = T_z^2$, $v \ge 0$, $v_0 > 0$ and $Av \le Cv$ componentwise (including $(Av)_i = 0$ where $v_i = 0$), then $W(2j, w) \le C^j$ for all j.*

Proof. $e_0 \le v/v_0$ and $A \ge 0$, so $A^j e_0 \le A^j v / v_0 \le C^j v / v_0$; take the entry 0. □

Proof of the upper bound. Let $K, L \to \infty$ with KL even. If K is odd then L is even and $Z(K,L) \le Z(K+1, L)$ (add a row of horizontal dominoes), so we may take $K = 2j$. Write $L = kw + r$, $0 \le r < w$. Cutting off a last strip of width r as in Lemma 4.6 gives $Z(K,L) \le W(K,w)^k W_r(K)$, where W_r counts the configurations of a width-r strip with weighted outside covers; crudely $W_r(K) \le (5 + 2z)^{Kr}$ (per cell: already covered, one of at most four later partners, or one of at most two outside covers). By Lemma 4.7,

$$\frac{\log Z(K,L)}{KL} \le \frac{k}{L}\frac{\log C_w}{2} + \frac{w \log(5+2z)}{L} \longrightarrow \frac{\log C_w}{2w}.$$

The certificate of §4.3 at $w = 24$ gives $\bar{s} \le 0.586216605404$. □

4.3. **Floating-point certificates.** The certificates use float64 vectors v produced by power iteration; how v was obtained does not matter. For the lower bound v is the normalised iterate $A^n e_0$; floating point never creates a positive entry from an exact zero (all terms are nonnegative), so $\operatorname{supp} v \subseteq \operatorname{supp} A^n e_0$. The code computes $y = \mathrm{fl}(Av)$, asserts $v_0 > 0$, $v_i > 10^{-280}$ on $\operatorname{supp} v$ and (upper bound) $y_i = 0$ off $\operatorname{supp} v$, and sets

$$c = \min_{i \in \operatorname{supp} v} \mathrm{fl}(y_i/v_i)\,(1 - 10^{-12}), \qquad C = \max_{i \in \operatorname{supp} v} \mathrm{fl}(y_i/v_i)\,(1 + 10^{-12}).$$

Each cell step writes every entry as a sum of at most six nonnegative terms (already covered, outside cover times z, right, down-left, down, down-right), so over the $2w$ steps of A, $\mathrm{fl}(Av)_i \in (Av)_i\,[(1-u)^{12w}, (1+u)^{12w}]$ with $u = 2^{-53}$: a relative error of at most $12wu\,(1+o(1)) = 3.2 \cdot 10^{-14}$

at $w = 24$, plus one rounding in the division. All weights are at least 1 (z is rounded up, so $T_{\mathrm{fl}(z)} \geq T_{\sqrt{\varphi}}$ entrywise) and the entries on the support exceed 10^{-280}, so there is no underflow and a float entry vanishes iff the exact one does. Since $10^{-12} \gg 3.2 \cdot 10^{-14} + u$, the inequalities $Av \geq cv$ and $Av \leq Cv$ hold exactly. The logarithms are taken with 30-digit arithmetic and the results are rounded outward to 12 decimals.

Audit. The following were checked in exact arithmetic: the cut identity and both strip inequalities on 58 boards ($K \leq 8$, $kw \leq 12$); $\max M(A, B) = F_{n+1}$ for all A, B in columns of length $m \leq 11$, and the branching step for $m \leq 8$; the transfer with weight z against brute force as a polynomial in z (162 evaluations); and, for $w = 2, \ldots, 12$, the float ratios against exact rational ratios for the same float vector ($|\mathrm{float}/\mathrm{exact} - 1| \leq 1.7 \cdot 10^{-16}$). The test has power: with $z = 0$ the strip inequality fails on all 40 boards with $k \geq 2$. On these small boards $z = 0.5$ would suffice; the weight $\sqrt{\varphi}$ is needed by the proof, not by the data.

5. Surface free energy

For the free strip of width k, $\log Z(K, k) = Ka_k + O(1)$, and

$$f(k) = a_k - sk \longrightarrow -2\sigma,$$

where σ is the free energy per unit length of a free edge. Even k approach the limit from above, odd k from below, and every even value lies above every odd one (Table 4; a_k from extended-precision power iteration for $k = 14, \ldots, 24$ and float64 with a Collatz–Wielandt bracket for $k = 25, 26$).

Table 4. $f(k) = \log \lambda_k - sk$ for free strips (`out_surface.txt`).

k	$f(k)$	k	$f(k)$
19	−0.4840358718678495	20	−0.4840358714180326
21	−0.4840358715709653	22	−0.4840358715195378
23	−0.4840358715369125	24	−0.4840358715310018
25	−0.4840358715330068	26	−0.4840358715323304

The odd sequence converges cleanly (all models of §3 agree to 10^{-16}); the even one carries two corrections, and only the two-exponential and Wynn models are stable under the backtest. The stable models give $-2\sigma \in [-0.4840358715325015, -0.4840358715325006]$; the error of s, multiplied by $k = 26$, dominates:

$$-2\sigma = -0.484035871532501 \pm 1.8 \cdot 10^{-14}, \qquad \sigma = 0.242017935766251 \pm 9 \cdot 10^{-15}.$$

The fit to the exact square boards in §3 has linear coefficient $a = -0.96808$ (four free edges), consistent with $-4\sigma = -0.9680717$.

6. Parity splitting and correlation length

For even w let $\Delta_w = \log \lambda_w^{(0)} - \log \lambda_w^{(1)}$ and $\xi_w = 2/\log(\Delta_{w-2}/\Delta_w)$ (Table 5). A pure exponential $\log \Delta_w = a - w/\xi$ fits $w = 10, \ldots, 24$ poorly ($\xi = 1.1931$, rms $1.5 \cdot 10^{-2}$); with a power prefactor, $\log \Delta_w = a - w/\xi + b \log w$ gives $\xi = 1.2323$, $b = -0.433$, rms $9.3 \cdot 10^{-4}$, and extrapolating $1/\xi_w$ linearly in $1/w$ gives $\xi_\infty = 1.2351$. So $\xi \approx 1.23$.

Inside a sector the transfer has a gap: at $w = 20$ ARPACK gives $\log(\lambda_1/|\lambda_3|) = 1.082727$ per row. The odd-width corrections behave like $e^{-w/\xi_{\mathrm{odd}}} w^b$ with $1/\xi_{\mathrm{odd}} \approx 1.07$, close to this gap. All finite-size corrections we see are exponentially small, as expected for a non-critical (gapped) dimer liquid. On the square lattice, which is critical, the corrections are algebraic, $(f_w - G/\pi)w^2 \to \pi/6$, and the same transfer-matrix extrapolation from $w \leq 18$ yields only $G/\pi \pm 6 \cdot 10^{-8}$.

TABLE 5. Splitting of the parity sectors on even cylinders (`out_estimate.txt`).

w	Δ_w	ξ_w	w	Δ_w	ξ_w
10	$1.521094 \cdot 10^{-4}$	1.15438	18	$1.788983 \cdot 10^{-7}$	1.19451
12	$2.782917 \cdot 10^{-5}$	1.17750	20	$3.373229 \cdot 10^{-8}$	1.19878
14	$5.124578 \cdot 10^{-6}$	1.18201	22	$6.389264 \cdot 10^{-9}$	1.20204
16	$9.544544 \cdot 10^{-7}$	1.19001	24	$1.214909 \cdot 10^{-9}$	1.20486

7. Checks on solvable lattices

The transfer code has a switch for the square lattice (no diagonals) and the triangular lattice (one diagonal). Against brute-force enumeration of perfect matchings it agrees on all 90 boards tested (king, triangular and square; free strips and cylinders; $w = 3, \dots, 6$, $h = 1, \dots, 4$). It reproduces the exact domicule counts of the repository for all square boards up to 18×18, for rectangles, and the 20×20 count.

Square lattice. The free-strip growth rates agree with the product formula $\prod(a + \sqrt{1 + a^2})$ and both cylinder sectors with the free-fermion formula to 10^{-16}.

Triangular lattice. The triangular lattice is planar, and we found a Pfaffian orientation by a sign pattern periodic modulo 2, checked against the exact counts on small boards and by $|\mathrm{Pf}| = Z$ on the 8×8, 10×10 and 12×12 boards (1848358752, 1852300296658080, 57261989986381118073728). Averaging the logarithm of the Bloch determinant over a 32×32 and a 64×64 grid of momenta gives the same value,

$$s_{\mathrm{tri}} = 0.428594537464959,$$

and the transfer matrix on the cylinder of circumference 17 gives 0.428594537464956, a difference of $2.9 \cdot 10^{-15}$. The triangular dimer model was solved in [4]; this check only tests our code.

8. A log-convexity conjecture

8.1. Statement and consequence.

Conjecture 8.1 (L)**.** *For every width $m \geq 1$ and every $q \geq 1$, $Z(2q, m)^2 \leq Z(2q-2, m)\, Z(2q+2, m)$.*

Only a much weaker statement about the strip growth rates is needed:

($\mathbf{L_\infty}$) $q \mapsto a_{2q} = \log \lambda_{2q}$ is convex on $2q \geq 22$, i.e. $a_{2q+2} - 2a_{2q} + a_{2q-2} \geq 0$ for $2q \geq 24$.

Proposition 8.2. *(L) implies (L_∞), and (L_∞) implies $\underline{s} \geq 0.565511911035$.*

Proof. Under (L), $q \mapsto m^{-1} \log Z(2q, m)$ is convex for each m; by $Z(2q, m) = Z(m, 2q)$ and (1), it tends to a_{2q} along even m, and a pointwise limit of convex sequences is convex. Under (L_∞) the increments $b_q = a_{2q+2} - a_{2q}$ are nondecreasing for $q \geq 11$; since $a_{2q}/(2q) \to s^*$, they increase to $2s^*$, so $s^* \geq b_{11}/2$, and $\underline{s} \geq s^*$ by the proof of the lower bound. Finally $b_{11} \geq \log \lambda_{24}^{\mathrm{low}} - \log \lambda_{22}^{\mathrm{up}}$, with the certified lower bound for $w = 24$ (§4.3, minus 10^{-14} for the float storage of the per-site value) and a certified upper bound $\log \lambda_{22} \leq 11.95722617139012$ from Lemma 4.7 with $z = 0$. This gives 0.565511911035. □

The gap between this bound and the estimate of s is $5.7 \cdot 10^{-12}$: all 132 candidates $(a_{2q+p} - a_{2q})/p$ with $2q \leq 22$, $2q + p \leq 24$ lie below s, the largest being 0.5655119110355926 ($2q = 22$, $p = 2$). This is the Calkin–Wilf form of the lower bound [2], whose standard proof via reflection positivity does not apply here (§8.4). The second differences of a_{2q} are positive and decay geometrically up to $2q = 24$: $4.69 \cdot 10^{-5}$, $3.99 \cdot 10^{-6}$, $5.53 \cdot 10^{-7}$, $5.63 \cdot 10^{-8}$, $6.96 \cdot 10^{-9}$, $7.62 \cdot 10^{-10}$, $9.0 \cdot 10^{-11}$,

$1.01 \cdot 10^{-11}$ at $2q = 10, 12, \ldots, 24$ (from uncertified power iteration, float error about 10^{-15}); so numerically (L_∞) holds at its first instance $2q = 24$, and beyond that it is extrapolation.

8.2. **Evidence.** Table 6 summarises the tests of (L): exact integer counts by a transfer modulo several 60-bit primes and the Chinese remainder theorem, and for $m = 25, 26$ an x87 computation with a rigorous rounding bound (each cell step writes a sum of at most five nonnegative terms, so the computed Z is within a factor $(1+u)^{5mk}$, $u = 2^{-64}$; the smallest certified margin is $1.237 \cdot 10^{-10}$). The margin $\log Z(2q-2) + \log Z(2q+2) - 2\log Z(2q)$ decreases monotonically in q for every m, by a ratio tending to 0.16–0.20; there is no dip at intermediate q and nothing that shrinks with m.

TABLE 6. Tests of (L), $Z(2q)^2 \le Z(2q-2)Z(2q+2)$; the row range shrinks as m grows (`out_ltest.txt`, `out_lcert_m25.txt`, `out_lcert_m26.txt`).

lattice, boundary	widths m	rows $2q$	cases	violations
king, free	2–24	≤ 120 to ≤ 20	873	0
king, cylinder	3–22	≤ 80 to ≤ 22	550	0
king, free (x87, certified)	25, 26	≤ 28	28	0
square, free / cylinder	≤ 12	≤ 60	319 / 290	0 / 0
triangular, cylinder	≤ 12	≤ 60	290	0
triangular, free	≤ 12	≤ 60	319	**92**

8.3. **Counterexamples in neighbouring models.** (L) is not a general property of dimer models. On free triangular strips it fails from $m = 5$ on: $Z(6,5) = 6475$, $Z(8,5) = 210560$, $Z(10,5) = 6842465$ and $210560^2 - 6475 \cdot 6842465 = 30552725 > 0$ (confirmed by brute force). On the king's graph with weights on the two diagonals (exact, $2q \le 38$): equal weights $x \in \{0, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 10, 100\}$ give no violation for $m \le 9$; the weights $(1, \frac{1}{2})$ fail from $m = 5$ on (14 of 19 values of q at $m = 5$), $(1, \frac{3}{4})$ fails at $m = 9$ (5 of 19, first at $2q = 30$), while $(1, \frac{9}{10})$ and $(1, \frac{19}{20})$ show no violation for $m \le 16$, $2q \le 78$. The violations sit mostly at odd m and large q, i.e. they come from an oscillating subleading eigenvalue, which we never saw with equal diagonal weights. A proof of (L) therefore cannot be uniform in the diagonal weights; the left–right symmetry of the king's graph is the natural ingredient. (L) is also specific to even row counts: for even m, $j \mapsto Z(2j+1, m)$ is log-*concave* in 217 of 220 tested cases.

8.4. **Obstructions to standard proofs.** *Reflection positivity* fails for every reflection line of the king lattice. Between rows a reflection maps one diagonal to the other, and the gluing form $M(A, B)$ (the crossing matchings of §4) is indefinite: its smallest eigenvalue is $1 - \sqrt{2}$ already for columns of length $m = 3$, and the eigenvector $(1, -\sqrt{2}, 1)$ on $A = \{0\}, \{1\}, \{2\}$ is realised by actual three-row half-boards (with weights $10, 4, 10$). Through a line of vertices the kernel is $[U \ne D]$ with eigenvalues ± 1. The transfer is not symmetric ($w = 3$: $T[\emptyset \to \{1\}] = 2$, $T[\{1\} \to \emptyset] = 0$). *Moment sequences.* If $Z(2q, m)$ were a Stieltjes moment sequence, (L) would follow; but e_0-visible eigenvalues of T^2 are non-real or negative from $m = 3$ on (small, at most $0.047\,\lambda_1^2$ in modulus at $m = 10$), and the Hankel matrices have negative minors. *Total nonnegativity* of the transfer fails for the same reason, and the Lindström–Gessel–Viennot lemma needs a planar path model that the crossing diagonals do not provide.

8.5. **Reformulations.** Superpose a matching X of rows $[0, 2q)$ with a matching Y of rows $[2, 2q+2)$ ($Z(2q)^2$ pairs), or X' of $[0, 2q+2)$ with Y' of $[2, 2q)$ ($Z(2q+2)Z(2q-2)$ pairs). The union H has degree one on the top rows $\{0, 1\}$ and the bottom rows $\{2q, 2q+1\}$ and degree two in between, so it consists of paths and even cycles. In the first case top–top and bottom–bottom paths have odd length and top–bottom (TB) paths even length; in the second all paths are odd. Each H has $2^{c(H)}$ labelings, $c(H)$ the number of cycles of length at least four, so

$$Z(2q-2)Z(2q+2) - Z(2q)^2 = \sum_{H:\, t(H)\geq 1} 2^{c(H)}\big([\text{all TB paths odd}] - [\text{all TB paths even}]\big),$$

where $t(H)$ is the number of TB paths. We checked this identity pair by pair for small (m, q) (up to $3.4 \cdot 10^6$ pairs). The odd configurations win separately for every t (e.g. $m = 5$, $q = 2$: $t = 2$: 1847660 against 912656; $t = 4$: 32616 against 0). So (L) asks for a sign-reversing injection that flips the parity of all top–bottom paths; by the counterexamples above it must use the geometry of the king lattice. Equivalently, cutting the board in the middle, (L) says that the indefinite form M is positive semidefinite on the span of two boundary vectors u_{q-1}, u_{q+1} of half-boards.

Open problem 8.3. Prove (L), or at least (L_∞), for the king's graph. Either would make the lower bound $s \geq 0.565511911035$ a theorem. Any proof must distinguish the king's graph from the triangular lattice and from the king's graph with unequal diagonal weights.

9. Reproducibility

All code and outputs are in the repository

`https://github.com/dawidrylko/domino-tiling`,

branch `d/king-entropy`, directory `research/king-entropy/`. The scripts need Python 3.13 with numpy 2.5.3, mpmath 1.4.1 and scipy 1.18.1, as pinned in `requirements.txt`. Every number in this paper is listed in `paper/NUMBERS.md`, with the committed output file it is read from and the command that wrote that file. The main commands, run from that directory:

command	output
`python3 tm2.py king cyl 3 13, ... 14 19`	`sec_king_cyl_*.json`
`python3 cylw.py king cyl` w p, $w = 20, \ldots, 25$	`eig_king_cyl_w*_p*.json`
`python3 estimate.py`	`out_estimate.txt`, `z2.json`
`python3 errbar.py`	`out_errbar.txt`
`python3 bounds.py 24`	`out_bounds.txt`, `bounds.json`
`python3 audit_bounds.py`	`out_audit_bounds.txt`
`python3 surface.py`	`out_surface.txt`
`python3 cw.py`	`out_cw.txt`
`python3 zexact.py ...`; `python3 ltest.py`	`zex_*.json`, `out_ltest.txt`
`python3 lcert.py 25 30, ... 26 30`	`out_lcert_m25.txt`, ...
`python3 lweight.py` [big]	`out_lweight*.txt`
`python3 validate.py`; `python3 crosscheck.py`	`out_validate.txt`, ...
`python3 paper/derived.py`	`paper/out_derived.txt`

The outputs were produced in a container with four cores and 15.7 GiB of memory. The widest runs dominate the cost: the cylinder $w = 25$ (8.4 GiB, 22.5 min), the certified bounds at $w = 24$ (22 min), and the certified (L) tests at $m = 26$ (8.0 GiB, 43 min). The script `repro.py` re-runs the first-round scripts from a clean export of the repository and compares their outputs byte for byte.

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